Questions

Q1.

A resistor is connected to a power supply.

The potential difference across the resistor is 6.0 V.

(i) Which of these corresponds to a potential difference of 6.0 V?

(1)

- ☐ **A** 6.0 joules per ohm
- ☐ **B** 6.0 amps per coulomb
- ☐ **C** 6.0 joules per coulomb
- ☐ **D** 6.0 amps per ohm

(ii) The resistor remains connected for a period of time.

The current in the resistor is 200 mA.
A total charge of 42 C flows through the resistor.
Calculate, in minutes, the time taken for this amount of charge to flow through the resistor.

(3)

time = ... minutes

(iii) Calculate the total energy transferred by the 6.0 V power supply when a charge of 42 C flows through the resistor.

(2)

energy = ... J

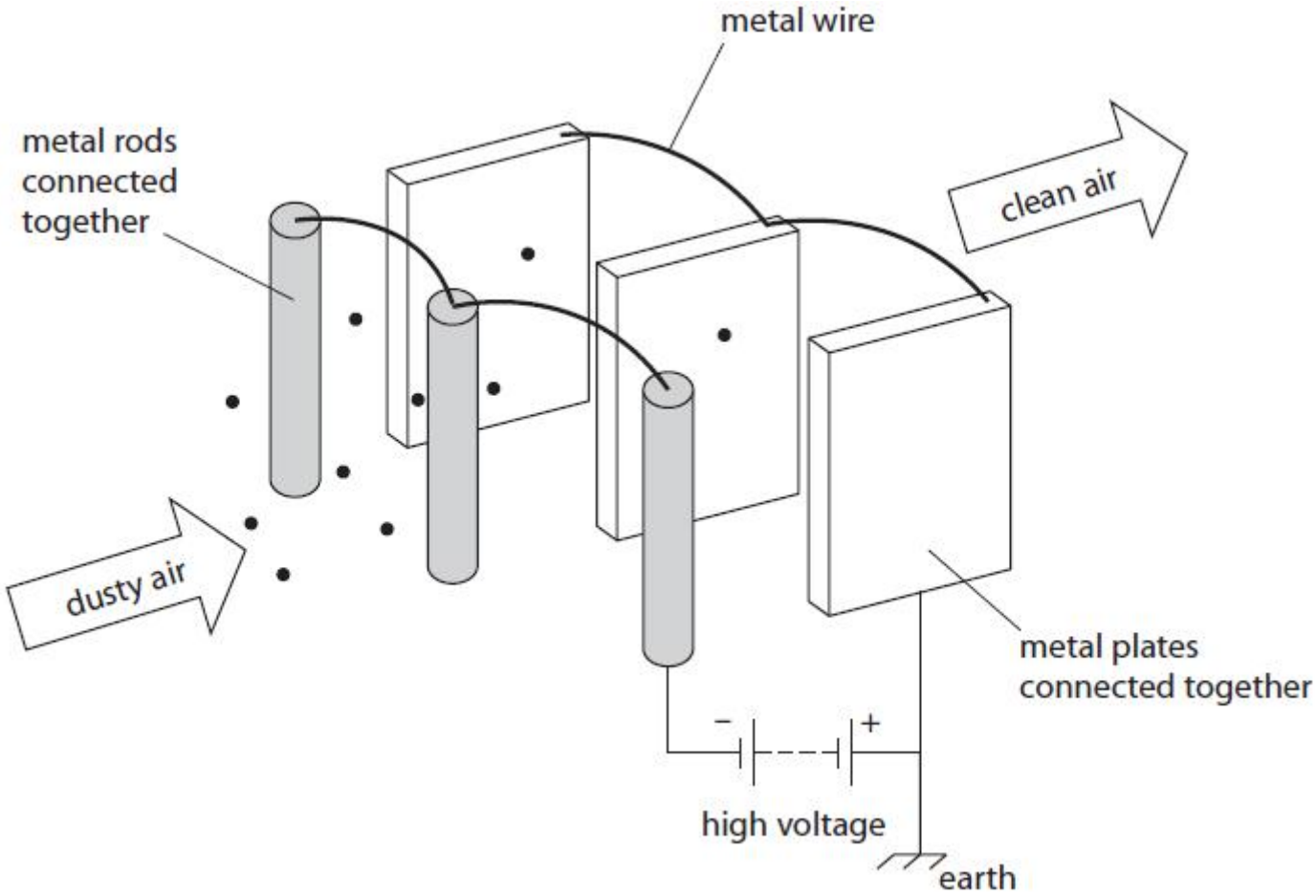
(Total for question = 6 marks)

Q2.

An electrostatic air filter is designed to remove dust particles from the air in a room.

A fan blows dusty air past several metal rods and metal plates.

There is a large potential difference (voltage) between the metal rods and the metal plates.



(a) Complete the sentence by putting a cross (☒) in the box next to your answer.

When dusty air goes past the metal rods, the dust particles become negatively charged.
This is because the dust particles

(1)

☐ **A** lose electrons

☐ **B** lose protons

☐ **C** gain electrons

☐ **D** gain protons

(b) When the dusty air flows past the metal plates, the dust particles settle on the metal plates.

Explain why the dust particles settle on the metal plates.

(2)

.....

.....

.....

.....

(c) (i) State what happens to the charge on the dust particles when they settle on the metal plates.

(1)

.....

.....

(ii) Explain why the charge does not build up on the metal plates.

(2)

.....

.....

.....

.....

(d) There is a current of 1.2 mA in the circuit.

Calculate the charge transferred by this current in 40 s.
State the unit.

(3)

charge transferred = ... unit: .....

Q3.

A man monitors how much money he spends on electricity.
He uses a device which calculates the cost of electrical energy used.
He connects his 2.9 kW electric kettle to the 230 V mains supply.

(i) Calculate the current in the kettle element.

(3)

current = ... A

(ii) The device shows that in one week the total cost of the electrical energy used by the kettle is 97 p.
1kW h of electrical energy costs 17 p.

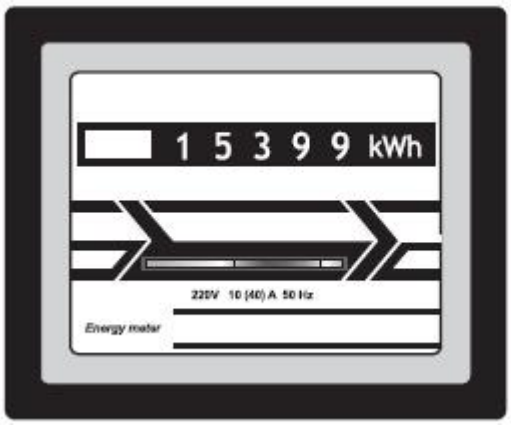
Calculate the length of time for which the kettle has been switched on during the week.

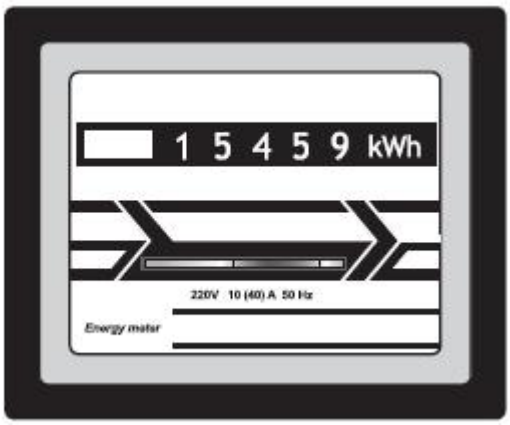
(3)

time = ... hours

Q4.

(a) Electricity costs 20p for each kW h.
The pictures show a domestic electricity meter at two different times.





(i) Calculate the cost of the electricity used between the two readings.

(2)

cost =.....p

(ii) The time between these two readings is 15 hours.
Calculate the average power supplied.

(2)

average power =.....kW

(b) Explain why step-up transformers are used in the transmission of electricity in the National Grid.

(2)

.....

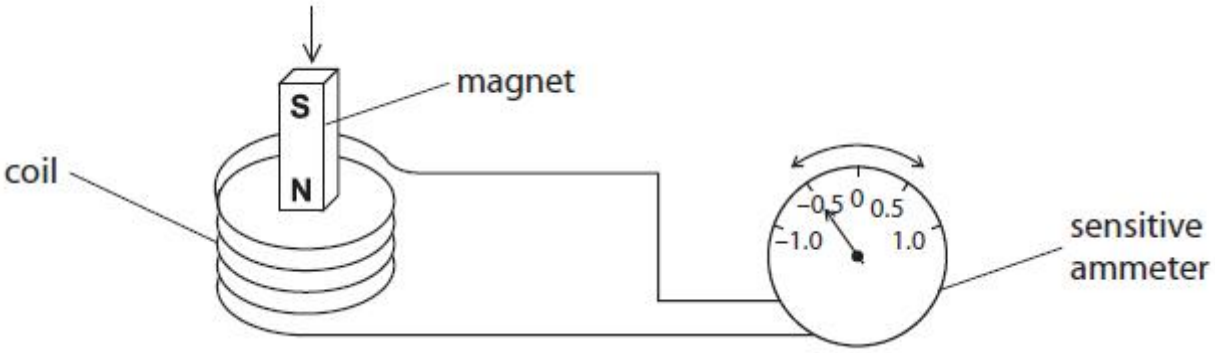
.....

.....

.....

*(c) The diagram shows a magnet moving into a coil of wire.

The coil of wire is attached to a sensitive ammeter.



The moving magnet and the coil of wire are producing an electric current.

The size and direction of the current can be changed in a number of ways.

Describe changes that can be made to produce different currents and the effect of each change.

(6)

.....

.....

.....

.....

.....

.....

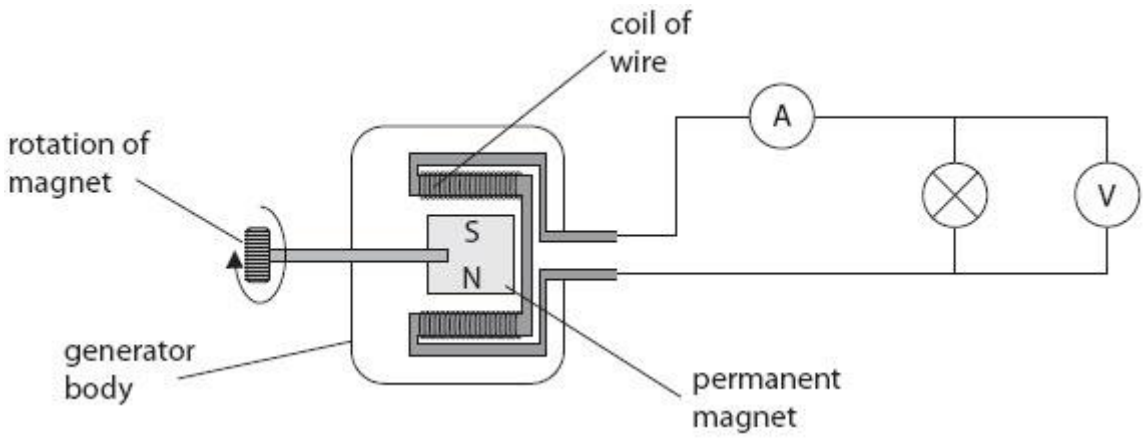
.....

.....

(Total for Question = 12 marks)

Q5.

The diagram shows a simple generator connected to a lamp.



The magnet is made to spin at a steady speed.
The ammeter gives a reading of 1.5 A.
The voltmeter gives a reading of 6 V.

(i) Calculate the output power of the generator.

(2)

.....

(ii) State two changes to the design of the generator that would give a larger output power for the same speed of rotation.

(2)

.....

.....

(iii) This generator supplies an alternating current (AC) to the lamp.
Other types of generators supply a direct current (DC).
Describe the difference between charge movement in a direct current and in an alternating current.

(2)

.....

.....

Q6.

Figure 3 shows some of the apparatus that students use to determine the resistance of a piece of iron wire.

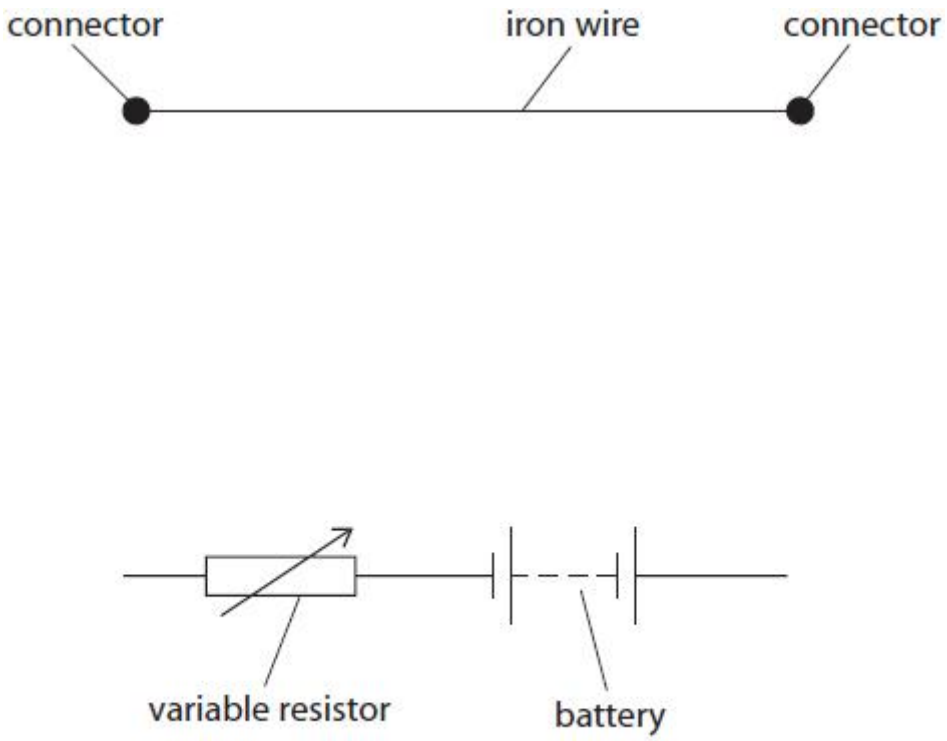


Figure 3

Add connecting wires, a voltmeter and an ammeter to complete the circuit in Figure 3 so that the students can determine the resistance of the piece of iron wire.

(Total for question = 2 marks)

Q7.

A student is given a low voltage power supply and 1 m of resistance wire.

The student uses these and other pieces of equipment to measure the resistance of just 50 cm of the resistance wire.

Draw a diagram of the circuit that the student should use.

Your circuit diagram should identify the pieces of equipment that the student uses.

(3)

(Total for question = 3 marks)

Q8.

Figure 4 shows the results from an experiment where the potential difference (voltage) across a filament lamp was varied.

The current and voltage were measured.

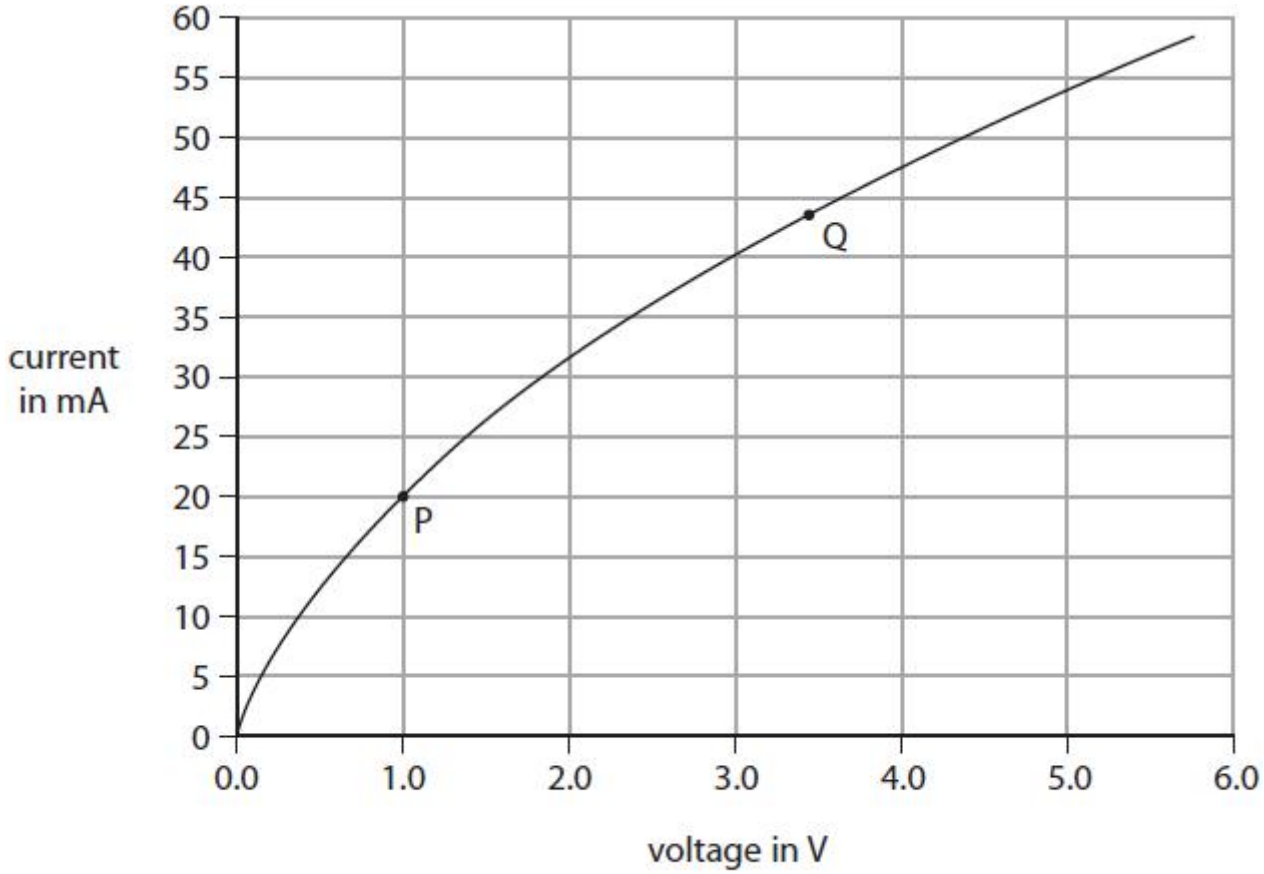


Figure 4

(i) Describe the relationship between the current and the voltage as shown in the graph in Figure 4.

(2)

.....

.....

.....

.....

(ii) Use the values of the voltage and current at point P and at point Q on the graph in Figure 4 to complete the table in Figure 5.

(2)

	voltage in V	current in mA
point P		
point Q		

Figure 5

(iii) Calculate the resistance of the filament lamp when the voltage is 4.5 V and the current is 51 mA.
Use the equation

$$R = \frac{V}{I}$$

(2)

resistance = ... Ω

(iv) Explain why the resistance of the filament lamp changes as the voltage across it increases.

(3)

.....

.....

.....

.....

.....

.....

(Total for question = 9 marks)

Q9.

A student investigates how the resistance of a thermistor varies with temperature.

Figure 16 shows a graph of the results of this investigation.

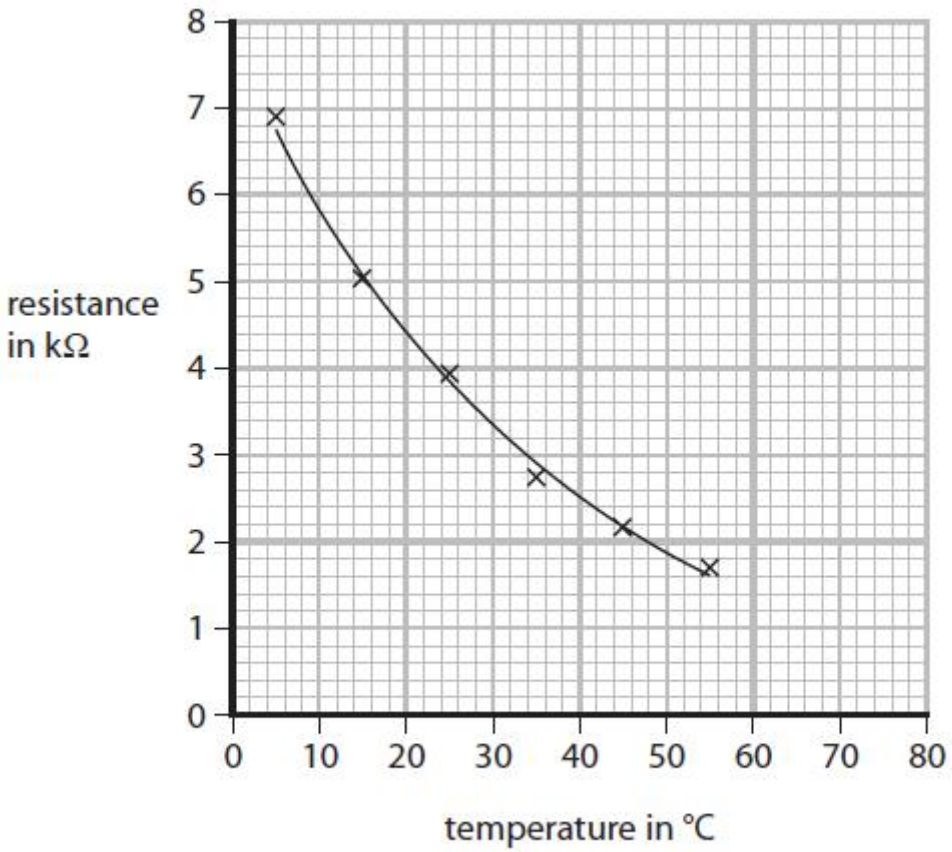


Figure 16

(i) Describe how the resistance of this thermistor varies with temperature.

(2)

.....

.....

.....

.....

(ii) Draw the tangent to the curve at a temperature of 30 °C, to find the rate of change of resistance with temperature at 30 °C.

State the unit.

(3)

rate of change of resistance with temperature at 30°C =..... unit .....

(Total for question = 5 marks)

Q10.

* The table in Figure 23 gives some information about **series** and **parallel** circuits.

series and **parallel** circuits
A battery can be connected to two equal value resistors in **series**. The same battery can be connected to the same two equal value resistors in **parallel**.
The current from the battery when the resistors are in **series** is different from the current from the battery when the resistors are in **parallel**.
The potential difference across each resistor when the resistors are in **series** will be different from the potential difference across each resistor when the resistors are in **parallel**.

Figure 23

Describe how a student can use ammeters and voltmeters to investigate the information given in Figure 23.

Your answer should include

- circuit diagrams showing how the currents and voltages can be measured
- a comparison of the results that would be expected

(6)

Space for circuit diagrams

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(Total for question = 6 marks)

Mark Scheme

Q1.

Question Number:	Answer	Mark
(i)	C 6.0 joules per coulomb **The only correct answer is C** ***A*** *is not correct because 1 volt is 1 joule per coulomb* ***B*** *is not correct because 1 volt is 1 joule per coulomb* ***D*** *is not correct because 1 volt is 1 joule per coulomb*	**(1)** AO 1 1

Question Number:	Answer	Additional Guidance	Mark
(ii)	recall and substitution (1) $42 = \frac{200 \times t}{(1000)}$ rearrangement (1) $t = \frac{42\ (\times 1000)}{200\ (\times 60)}$ evaluation (1) (t =) 3.5 (minutes)	accept substitution and rearrangement in either order 2.1 to any power of 10 or 3.5 to any power of 10 scores 2 marks 3 minutes 30 seconds award full marks for correct answer without working	**(3)** AO 1 1 AO 2 1

Question Number:	Answer	Additional Guidance	Mark
(iii)	recall and substitution (1) (E =) 42 x 6.0 evaluation (1) (energy =) 250 (J)	(using E = VIt) (E =) 6.0 x 200 (x 10^{-3}) x 2.10 (x 10^{2}) accept 252 (J) award full marks for correct answer without working	**(2)** AO 1 1 AO 2 1

Q2.

Question Number	Answer	Acceptable answers	Mark
(a)	C (gain electrons)		**(1)**

Question Number	Answer	Acceptable answers	Mark
(b)	An explanation linking • (Force of) attraction (1) • (plates have) opposite charge (to dust) (1)	 Plates have a positive charge Ignore different charge	**(2)**

Question Number	Answer	Acceptable answers	Mark
(c)(i)	transferred to plate / lost (1)	neutral / become discharged	**(1)**

Question Number	Answer	Acceptable answers	Mark
(c)(ii)	An explanation linking any two of • Metal is a conductor (1) • Electrons / (negative) charge moves (through the plates/ wire) (1) • Towards the voltage supply / earth /ground (1)	 Metal not an insulator Plates / charges are earthed	**(2)**

Question Number	Answer	Acceptable answers	Mark
(d)	Substitution: $Q = 1.2 \times 10^{-3} \times 40$ (1) Evaluation: 0.048 or 4.8×10^{-2} (1) C / coulombs (1)	Give 2 marks for correct answer with no working shown Unit mark is independent Allow for 1 mark 48 (with incorrect or no units) Allow for 2 marks 48 C Allow for all 3 marks 48 mC	**(3)**

Q3.

Question Number	Answer	Acceptable answers	Mark
(i)	Substitution (1) 2900 = 230 × current Transposition (1) $\frac{2900}{230}$ Evaluation (1) 13 (A)	Award full marks for correct answer with no working Allow substitution and transposition in either order Ignore powers of ten errors until evaluation Allow numbers which round up to 13	**(3)**

Question Number	Answer	Acceptable answers	Mark
(ii)	Substitution (1) 97 = 2.9 × time × 17 Transposition (1) $\frac{97}{2.9 \times 17}$ OR $\frac{97}{49.3}$ Evaluation (1) 2.0 (h)	Award full marks for correct answer with no working Allow substitution and transposition in either order Ignore powers of ten errors until evaluation Allow $\frac{97}{17}$ = 5.7 for 1 mark Allow numbers which round up to 2.0	**(3)**

Q4.

Question Number	Answer	Acceptable answers	Mark
(a)(i)	60 (kW h/ units) (1) 60 x 20 (= 1200) (p) (1)	15459 - 15399 £12 ecf Award full marks for correct answer with no working £12 scores 2 Power of Ten error scores maximum 1 60 in answer space with no working scores 1	**(2)**

Question Number	Answer	Acceptable answers	Mark
(a)(ii)	60 / 15 (1) 4 (kW) (1)	Allow ecf from 6(a)(i) marking point 1 Award full marks for correct answer with no working	**(2)**

Question Number	Answer	Acceptable answers	Mark
(b)	An explanation linking any two of: • increase voltage (1) • decrease current (1) • reduce {loss / waste} of {energy / heat} (1)	Increase efficiency (of energy transmission) Ignore "more efficient" by itself Accept power instead of energy Accept no energy loss	**(2)**

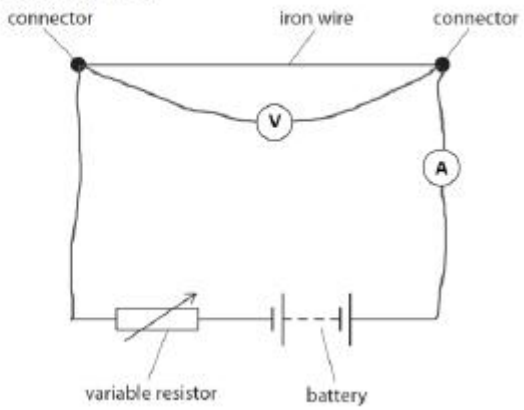
Question Number		Indicative content	Mark
QWC	*** (c)**	A description to include some of the following points • speed of movement • stronger / more powerful (ORA) magnet • more turns / coils (ORA) • iron core • reversing movement • turning the magnet round • effect of any / each change • more conducting / less resistant wire • allow stronger current • allow ammeter reading / recording / voltage for current • allow moving coil Correct ideas but using inaccurate scientific terminology • larger / bigger magnet • more / longer movement Ignore • irrelevant information • speeds up current or more electricity	**(6)**
Level	**0**	no rewardable material	
1	**1-2**	• a limited description of any one change e.g. use more coils **OR** a stronger magnet. • the answer communicates ideas using simple language and uses limited scientific terminology • spelling, punctuation and grammar are used with limited accuracy	
2	**3-4**	• a simple description of any two different changes **OR** one change and its effect e.g. use more coils and a weaker magnet **OR** more coils more current • the answer communicates ideas showing some evidence of clarity and organisation and uses scientific terminology appropriately • spelling, punctuation and grammar are used with some accuracy	
3	**5 - 6**	• a detailed description of a change linked to its effect and a second different change e.g. using more turns of wire makes a bigger current. Moving the magnet out. • the answer communicates ideas clearly and coherently uses a range of scientific terminology accurately • spelling, punctuation and grammar are used with few errors	

Q5.

	Answer	Acceptable answers	Mark
(i)	Substitution (1) 1.5 × 6 Evaluation (1) 9 (W) Ignore any unit given by	Power of 10 error max 1 mark Give full marks for correct answer with no working shown	**(2)**

	candidate.		
(ii)	• More turns on the coil (1) • More powerful/stronger magnet(s) (1)	Wrap coils on iron (core/former)/ more coils/twists/loops. Bigger coil is insufficient. More magnets. Bigger/larger magnet is insufficient. Ignore increase speed of rotation	**(2)**
(iii)	A description including • in one direction only for DC (1) • reversing direction for AC (1)	'DC goes straight' is insufficient AC switches/changes direction OR moves to and fro 'AC goes different ways' is insufficient. Diagram with labelled arrows could get 2 marks.	**(2)**

Q6.

Question number	Answer	Additional guidance	Mark
	voltmeter connected in parallel with the iron wire / any part of the iron wire (1) ammeter connected in series with the iron wire (1) example:  	accept any recognisable symbols. accept symbol drawn over connecting wire do not credit the same type of meter shown in contradictory positions	**(2)** **AO1**

Q7.

Question number	Answer	Additional guidance	Mark
	a diagram of a circuit including **all** of the following: power supply / cell(s) / battery identifiable resistance wire an ammeter a voltmeter (1) plus any **two** from ammeter in series (1) voltmeter in parallel (1) indication of tapping off / using 50cm of resistance wire (1)	accept symbols accept ohmmeter with resistance wire only ignore lamp(s) / additional resistors allow ohmmeter (across wire) instead of ammeter and voltmeter for 1 mark e.g. (crocodile) clips	**(3)** **AO2.2**

Q8.

Question number	Answer	Additional guidance	Mark
(i)	A description including as the potential difference (voltage) increases so does the current (1) idea of gradient of graph decreasing as V increases (1)	 positive correlation at a decreasing rate non-linear not directly proportional	**(2)** **AO3**

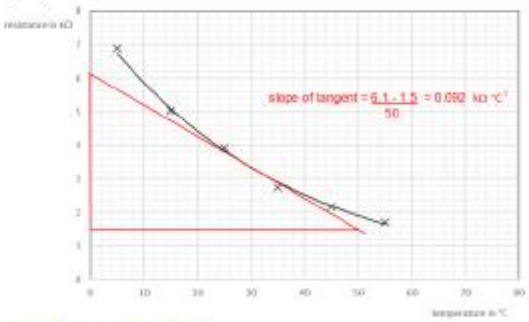
Question number	Answer	Additional guidance	Mark
(ii)	Award one mark for each row of the table voltage in V / current in mA point P: 1(.00) / 20 point Q: 3.4 ±0.1 / 43 ±1	ignore any units added in the boxes	(2) AO2

Question number	Answer	Additional guidance	Mark
(iii)	substitution (1) $(R=)\ \frac{4.5}{51(\times 10^{-3})}$ evaluation (1) 88.(2) (Ω)	0.088(2) or 8.8(2) or 0.88(2) or 0.09 seen scores 1 mark 0.088(2) kΩ or 0.09 kΩ scores 2 marks award full marks for correct answer without working	(2) AO2

Question number	Answer	Additional guidance	Mark
(iv)	an explanation linking any three of: identification of resistance increasing (1) heating (of the filament) (1) because of more collisions (1) of electrons (with ions / atoms / other electrons) (1)	temperature increases	(3) AO1

Q9.

Question number	Answer	Additional guidance	Mark
(i)	A description to include as temperature increases resistance decreases (1) non-linear / decreasing gradient (1)	 ORA allow exponential / inversely proportional in this context curve gets less steep as temperature increases ignore negative correlation unqualified quoted values are insufficient	(2) AO3.1

Question number	Answer	Additional guidance	Mark
(ii)	uses a right-angled triangle to calculate slope with a line of grazing incidence at θ = 30° C (1)  evaluation (1) (-) 0.092 (kΩ / °C) unit (1) kΩ / °C or kΩ °C^{-1}	tangent seen and used, drawn between θ = 25 and 35 °C accept for 2 marks either between 0.087 and 0.097 (kΩ / °C) or between 87 and 97 (Ω / °C) kohm/K or kohm K^{-1}	(3) AO3.2

Q10.

Question number	Indicative content	Mark
*	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. Description (may be inferred from circuit diagram(s)) • Connect ammeter in series with resistor(s) • Connect voltmeter across / in parallel with resistor(s) • Move meters to new positions. Circuit diagram(s) (accept any recognisable symbol) • Two resistors in series with a battery • Two resistors in parallel with a battery • An ammeter in series with the resistor(s) and battery • A voltmeter in parallel with the resistor(s) Expected results (allow reverse arguments) • Current (from battery) greater in parallel circuit than in series circuit • Pd across resistor(s) less in series circuit than in parallel circuit • Pd across resistor(s) in parallel circuit same as pd across battery • Pd across battery shared between resistors in series • Current (from battery) shared between resistors in parallel • Current same throughout whole of series circuit	**(6)** **AO1**

Level	Mark	Descriptor
	0	• No rewardable material.
Level 1	1-2	• Demonstrates elements of physics understanding, some of which is inaccurate. Understanding of scientific, enquiry, techniques and procedures lacks detail. (AO1) • Presents a description which is not logically ordered and with significant gaps. (AO1)
Level 2	3-4	• Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas, enquiry, techniques and procedures is not fully detailed and/or developed. (AO1) • Presents a description of the procedure that has a structure which is mostly clear, coherent and logical with minor steps missing. (AO1)
Level 3	5-6	• Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas, enquiry, techniques and procedures is detailed and fully developed. (AO1) • Presents a description that has a well-developed structure which is clear, coherent and logical. (AO1)

Level	Mark	Additional Guidance	General additional guidance – the decision within levels e.g. - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level.
	0	No rewardable material.	
Level 1	1–2	Additional guidance description of that would allow at least two measurements **or** an expected result of those measurements	Possible candidate responses any two correctly connected meters (may be in different circuits) ***or*** resistors in a series and a parallel circuit ***or*** current in parallel circuit shared between resistors ***or*** Pd of supply shared between resistors in series
Level 2	3–4	Additional guidance descriptions of obtaining two measurements ***with*** an expected results of those measurements for at least one circuit **or** expected results of those comparisons between circuits **without** description of obtaining measurements	Possible candidate responses a series and a parallel circuit with a correctly connected meter ***with*** current in parallel circuit shared between resistors but pd across them is same as supply **or** (total) current more in parallel than in series circuit and pd across each resistor in series less than in parallel **without** creditworthy circuit diagram or description of obtaining measurements

Level 3	5–6	Additional guidance descriptions of obtaining comparable measurements ***with*** expected results of those comparisons between circuits	Possible candidate responses a series and a parallel circuit with at least one correctly connected ammeter and voltmeter ***with*** (total) current more in parallel than in series circuit and pd across each resistor in series less than in parallel